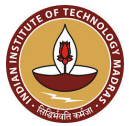
Case IV: Non-homogeneous Boundary Conditions

- Problem. A wire AB of length L has its ends A and B maintained at A_0° and B_0° respectively until the steady-state condition is reached.
- Then the temperatures at A and B are suddenly changed to A_1° and B_1° and maintained thereafter.
- Find the subsequent temperature distribution in the wire.
- The boundary conditions $u(0,t) = A_1$, $u(L,t) = B_1$ are non-homogeneous (non-zero).
- Separation of variables cannot be applied directly: a sum of solutions would not satisfy them.
- Plan: (I) find the initial temperature, (II) find the new steady state u_s, (III) find the transient $u_{tr} = u - u_s$, which has zero boundary values.



Step I: The Initial Temperature

- In the steady state u is independent of t, i.e. $u_t = 0$.
- The heat equation then reduces to

$$\frac{d^2u}{dx^2} = 0 \quad \Rightarrow \quad u(x) = ax + b.$$

- The boundary conditions at the old steady state are $u(0) = A_0$, $u(L) = B_0$.
- Hence $b = A_0$ and $aL + A_0 = B_0$, so $a = \dfrac{B_0 - A_0}{L}$.
- This old steady state is the initial temperature for the new problem:

$$u(x, 0) = \frac{B_0 - A_0}{L}\, x + A_0. \tag{1}$$

- A steady-state temperature in a bar without sources is always linear in x.



Step II: The New Steady State

- Now the boundary conditions are changed to

$$u(0,t) = A_1, \qquad u(L,t) = B_1. \tag{2}$$

- We have to solve $u_t = k^2 u_{xx}$ (3) with the initial condition (1) and boundary conditions (2).
- We assume the solution in the form

$$u(x,t) = u_s(x) + u_{tr}(x,t), \tag{4}$$

where u_s is the steady-state solution with the new boundary values, and u_{tr} is the transient solution.

- $u_s'' = 0$ with $u_s(0) = A_1$, $u_s(L) = B_1$ gives $u_s = Cx + D$, $D = A_1$, $CL + A_1 = B_1$:

$$u_s(x) = \frac{B_1 - A_1}{L}\, x + A_1. \tag{5}$$



Step III: The Transient Problem

- From (4), $u_{tr}(x,t) = u(x,t) - u_s(x)$, and u_{tr} also satisfies the heat equation (since $u_s'' = 0$).
- Boundary conditions:

$$u_{tr}(0,t) = A_1 - A_1 = 0, \qquad u_{tr}(L,t) = B_1 - B_1 = 0. \tag{6}$$

- So the boundary conditions are homogeneous for u_{tr}.
- Initial condition, by (1) and (5):

$$u_{tr}(x,0) = \left(\frac{B_0 - A_0}{L}x + A_0\right) - \left(\frac{B_1 - A_1}{L}x + A_1\right) = A_2\frac{x}{L} + B_2 =: f(x), \tag{7}$$

with $A_2 = (B_0 - A_0) - (B_1 - A_1) = (A_1 - A_0) + (B_0 - B_1)$ and $B_2 = A_0 - A_1$.



Step III (contd.): Solving for u_{tr}

- $(u_{tr})_t = k^2 (u_{tr})_{xx}$ with (6) and (7) is exactly Case I, so

$$u_{tr}(x,t) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{L}\, e^{-\lambda_n^2 t}, \qquad B_n = \frac{2}{L}\int_0^L f(x) \sin \frac{n\pi x}{L}\, dx.$$

- With $f(x) = A_2 \frac{x}{L} + B_2$, integration by parts gives

$$B_n = \frac{2}{L}\Big[-\Big(A_2 \frac{x}{L} + B_2\Big)\frac{L}{n\pi} \cos \frac{n\pi x}{L} + \frac{A_2}{L}\frac{L^2}{n^2\pi^2} \sin \frac{n\pi x}{L}\Big]_0^L.$$

- The sine terms vanish at both limits, so

$$B_n = -\frac{2}{n\pi}\big[(A_2 + B_2)\cos n\pi - B_2\big] = \frac{2}{n\pi}\big[B_2 - (-1)^n (A_2 + B_2)\big].$$



The Complete Solution

- Therefore the required temperature distribution is $u = u_s + u_{tr}$:

$$u(x,t) = (B_1 - A_1)\frac{x}{L} + A_1 + \frac{2}{\pi}\sum_{n=1}^{\infty} \frac{B_2 - (-1)^n(A_2 + B_2)}{n} \sin\frac{n\pi x}{L}\, e^{-n^2\pi^2 k^2}$$

  where $A_2 = (B_0 - A_0) - (B_1 - A_1)$ and $B_2 = A_0 - A_1$.
- Note $A_2 + B_2 = B_0 - B_1$: the change of temperature at the end B.
- $B_2 = A_0 - A_1$ is the change of temperature at the end A (with sign reversed).
- As $t \to \infty$, $u_{tr} \to 0$ and $u \to u_s$: the new linear steady state.
- The same idea (subtract a function satisfying the boundary conditions) works for other non-homogeneous conditions too.



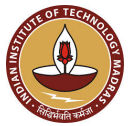
Example 7: A Bar of Length 10 cm

- A bar AB of length 10 cm has its ends A and B kept at 30° and 100° respectively until a steady state is reached.
- Then the temperature at A is lowered to 20° and that at B to 40°, and these are maintained. Find the subsequent temperature distribution.
- Step 1: with $u_t = 0$, $u = ax + b$; $u(0) = 30$, $u(10) = 100$ give $b = 30$, $a = 7$:

$$u(x, 0) = 7x + 30.$$

- New boundary conditions: $u(0, t) = 20$, $u(10, t) = 40$ for all t.
- Step 2: $u_s(x) = a_1 x + b_1$ with $u_s(0) = 20$, $u_s(10) = 40$: $b_1 = 20$, $a_1 = \dfrac{40 - 20}{10} = 2$.

$$u_s(x) = 2x + 20.$$



Example 7 (contd.)

- Step 3: $u_{tr} = u - u_s$ satisfies $u_{tr}(0,t) = 0 = u_{tr}(10,t)$ and

$$u_{tr}(x,0) = (7x + 30) - (2x + 20) = 5x + 10 = f(x).$$

- So $u_{tr}(x,t) = \sum_{n=1}^{\infty} B_n \sin \dfrac{n\pi x}{10}\, e^{-k^2 n^2 \pi^2 t/100}$ with

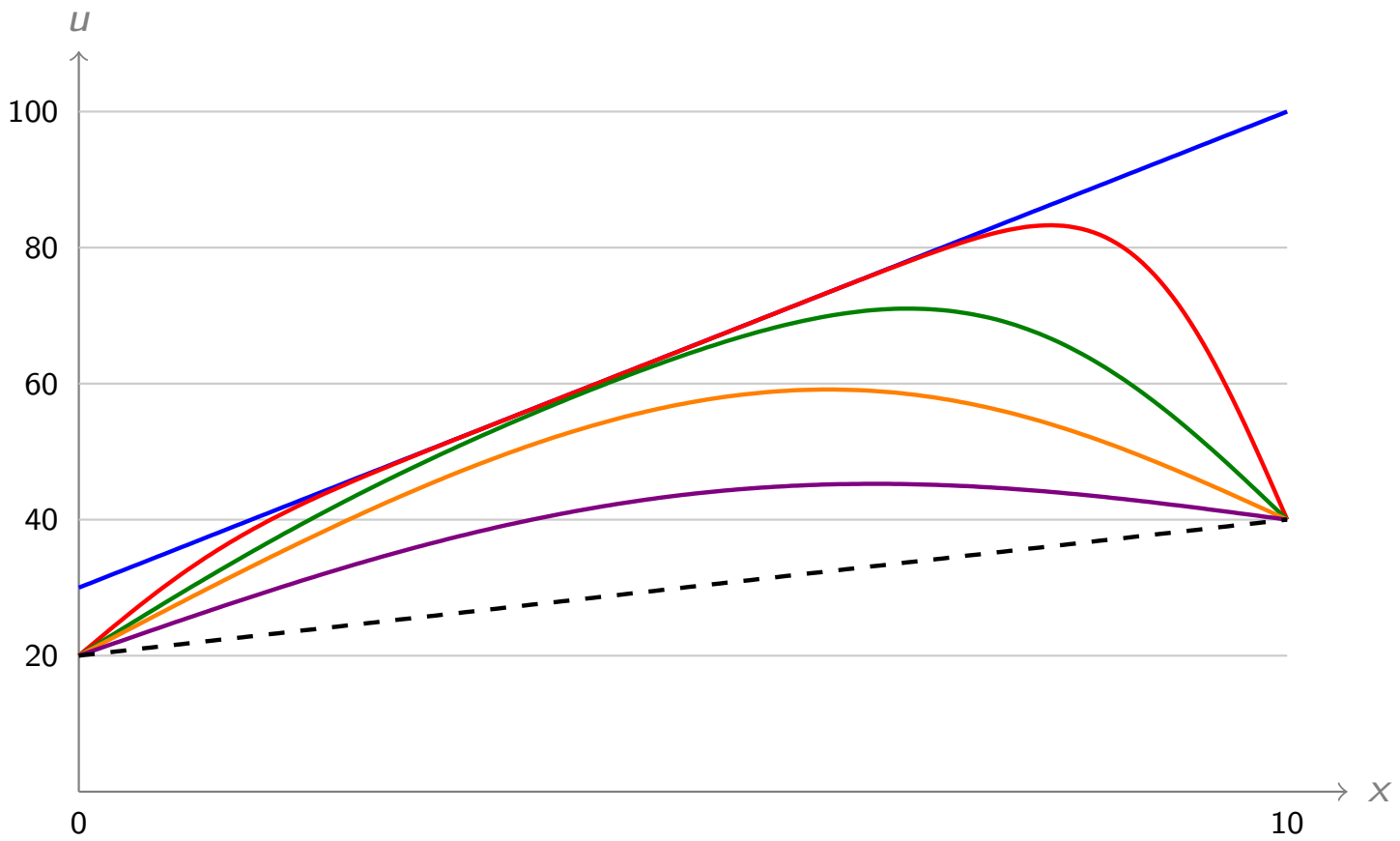
$$B_n = \frac{2}{10}\int_0^{10} (5x + 10) \sin \frac{n\pi x}{10}\, dx = \frac{20}{n\pi}\left[1 - 6(-1)^n\right].$$

- (Check with the general formula: $B_2 = 10$, $A_2 + B_2 = 60$.).
- The desired solution is

$$u(x,t) = 2x + 20 + \frac{20}{\pi}\sum_{n=1}^{\infty} \frac{1 - 6(-1)^n}{n} \sin \frac{n\pi x}{10}\, e^{-k^2 n^2 \pi^2 t/100}.$$

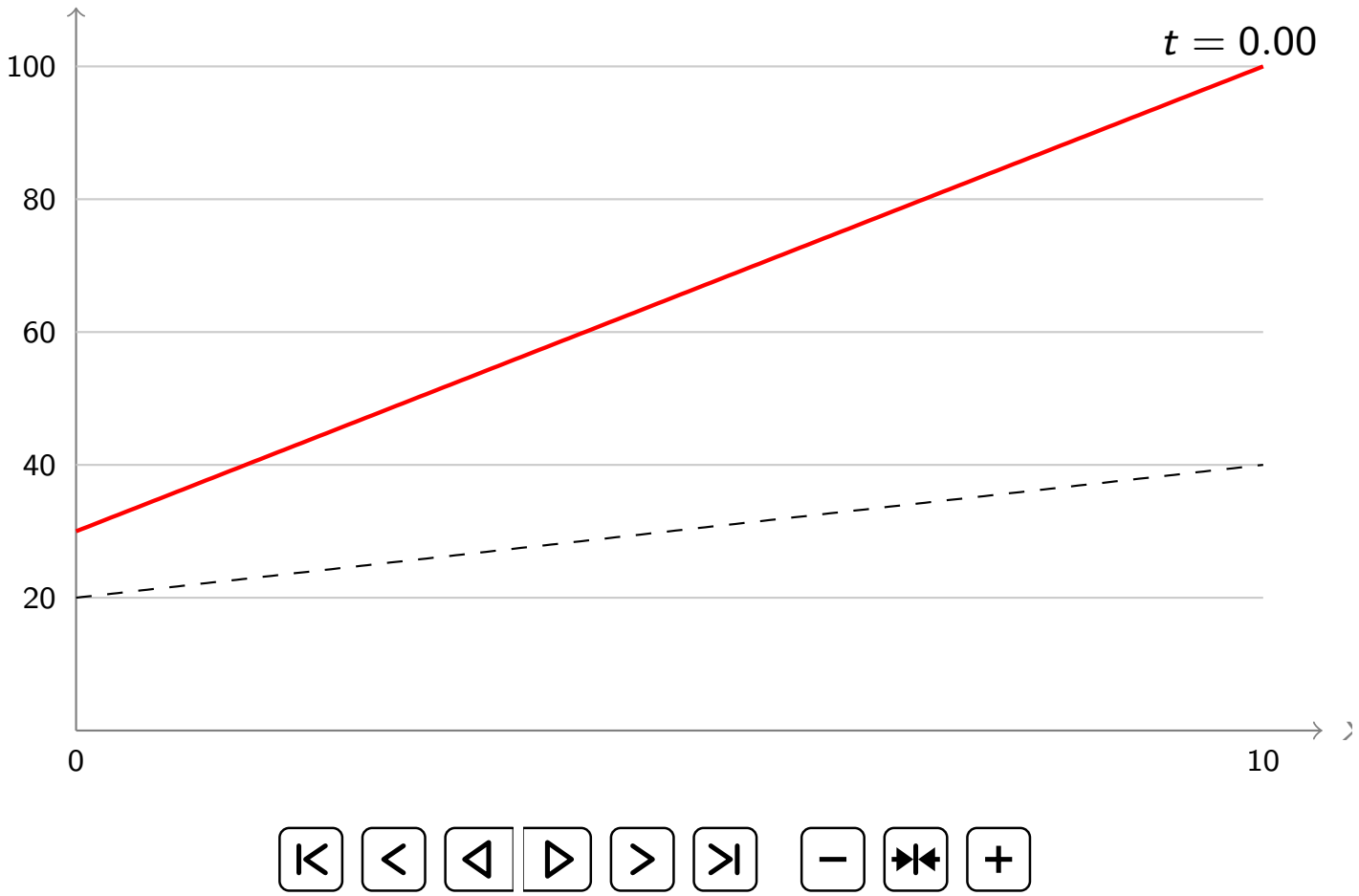


Picture: From One Steady State to Another



- Initial line $7x + 30$ (blue), profiles at $t = 0.5, 2, 5, 12$ ($k = 1$), and the final line $2x + 20$ (dashed).

Animation: Example 7



- The ends jump to the new values at once; the interior relaxes to the new line $2x + 20$ (dashed).
- Animation plays in Adobe Reader or Okular; time is shown on a logarithmic scale.

Homework (Case IV)

- (1) Bar length $L = 20$; end temperatures 0° and 100° (steady state), then changed to 25° and 75°.

  Ans: $u(x,t) = \frac{5}{2}x + 25 - \frac{50}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\sin\frac{n\pi x}{10}\,e^{-k^2n^2\pi^2t/100}$.

- (2) Bar length $L = 10$; end temperatures 20° and 40° (steady state), then changed to 50° and 10°. Show that the temperature at the middle point remains unaltered for all time.

  Ans: $u(x,t) = 50 - 4x - \frac{120}{\pi}\sum_{n=1}^{\infty}\frac{1}{2n}\sin\frac{2n\pi x}{10}\,e^{-4k^2n^2\pi^2t/100}$;

  $u(5,t) = 30$.



Homework (Case IV, contd.)

- (3) Solve $u_t = k^2 u_{xx}$, $0 \le x \le 1$, $u(0,t) = 2$, $u(1,t) = 3$, $u(x,0) = x(1-x)$.

  Hint: $u_s = x + 2$, $u_{tr}(x,0) = -(x^2+2)$.

  Ans:

$$u = x + 2 + 2\sum_{n=1}^{\infty} \left[\frac{3(-1)^n - 2}{n\pi} + \frac{2\left(1-(-1)^n\right)}{n^3\pi^3} \right] \sin n\pi x \, e^{-k^2 n^2 \pi^2 t}.$$

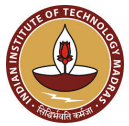


Case V: Steady-state Initial Data, Zero End Temperatures

- Example 8. A bar of length L, laterally insulated, has its ends A and B kept at 0° and u_0° respectively until the steady state prevails.
- The temperature at B is then suddenly reduced to 0° and kept so, while A is maintained at 0°. Find the temperature distribution.
- Initial temperature from the steady state ($u_{xx} = 0$, $u = ax + b$): $u(0) = 0 \Rightarrow b = 0$, $u(L) = u_0 \Rightarrow a = \frac{u_0}{L}$.

$$u(x, 0) = \frac{u_0}{L}\, x, \qquad u(0, t) = 0, \quad u(L, t) = 0.$$

- Now the boundary conditions are homogeneous, so this is Case I with $f(x) = \frac{u_0}{L} x$ (here $u_s \equiv 0$).



Example 8 (contd.): Solution

- $u(x,t) = \sum_{n=1}^{\infty} B_n \sin \dfrac{n\pi x}{L}\, e^{-n^2\pi^2 k^2 t/L^2}$, with

$$B_n = \frac{2}{L}\int_0^L \frac{u_0 x}{L} \sin \frac{n\pi x}{L}\, dx = \frac{2u_0}{n\pi}(-1)^{n+1}.$$

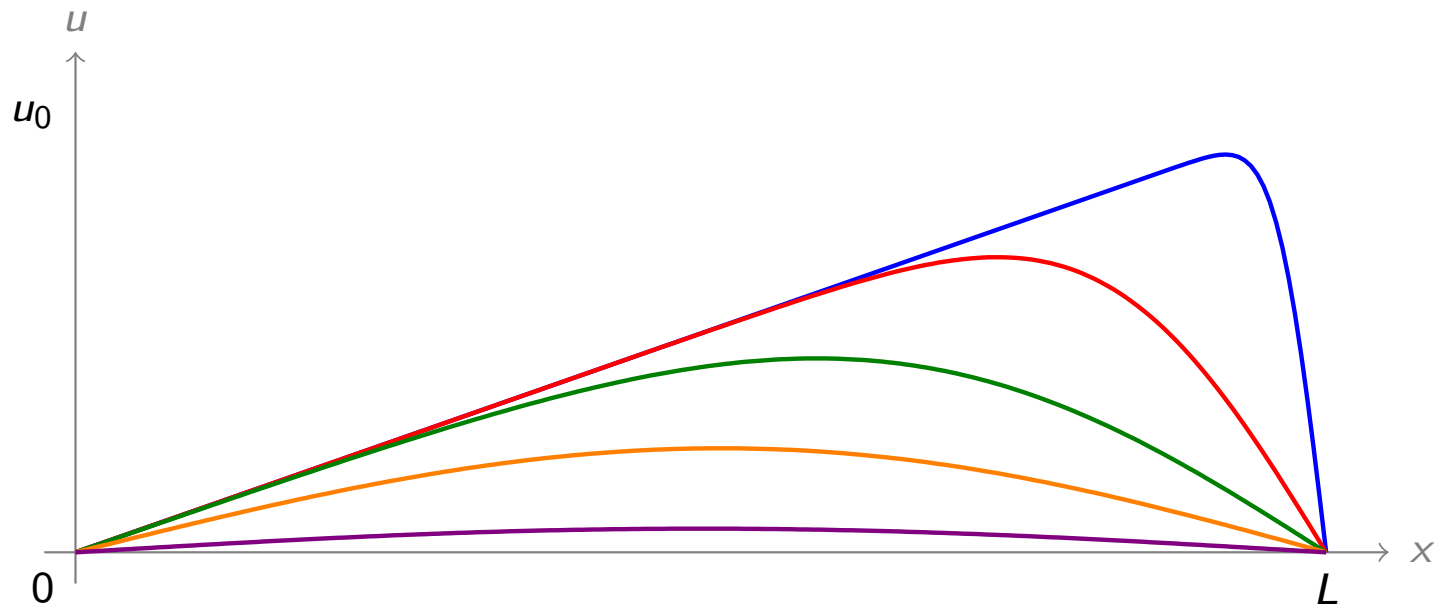
- Hence

$$u(x,t) = \frac{2u_0}{\pi}\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin \frac{n\pi x}{L}\, e^{-n^2\pi^2 k^2 t/L^2}.$$

- At $t = 0$ this is the sine series of $u_0 x/L$; it converges to $u_0 x/L$ for $0 \le x < L$, and to 0 at $x = L$.
- Coefficients decay only like $1/n$, reflecting the jump at $x = L$; for $t > 0$ the exponentials make the series converge rapidly.
- This is also the special case $A_0 = A_1 = B_1 = 0$, $B_0 = u_0$ of Case IV.



Picture: Example 8 ($u_0 = 1$, $L = 1$, $k = 1$)



- $t = 0.0005, 0.01, 0.04, 0.1, 0.25$: the hot end B is cooled suddenly, and heat flows out through it.
- The maximum moves inside the bar and decays; eventually $u \approx \frac{2u_0}{\pi} \sin \frac{\pi x}{L} e^{-\pi^2 k^2 t / L^2}$.

Homework (Case V)

- (1) A bar of 30 cm length has its ends kept at 20° and 80° respectively until the steady-state condition prevails. The temperature of each end is then suddenly reduced to 0° and maintained thereafter. Find the temperature in the bar.

  Ans: $u(x,t) = \dfrac{40}{\pi} \displaystyle\sum_{n=1}^{\infty} \dfrac{1-4(-1)^n}{n} \sin \dfrac{n\pi x}{30}\, e^{-k^2 n^2 \pi^2 t/900}$.



Homework (Case V, contd.)

- (2) A rod of 100 cm length has its ends kept at 0° and 100° until the steady state prevails. The two ends are then suddenly insulated and kept so. Find the temperature in the rod, and show that the sum of the temperatures at any two points equidistant from the centre is always 100°.

  Hint: $u(x,0) = x$; solve with $u_x(0,t) = u_x(100,t) = 0$.

  Ans: $u = 50 - \dfrac{400}{\pi^2} \displaystyle\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos \frac{(2n-1)\pi x}{100}\, e^{-k^2(2n-1)^2\pi^2 t/10^4}$.

  Since $\cos \frac{(2n-1)\pi(100-x)}{100} = -\cos \frac{(2n-1)\pi x}{100}$, we get $u(x,t) + u(100-x,t) = 100$.



Radiation (Robin) Boundary Condition

- Suppose the end $x = 0$ is kept at 0° and heat radiates from the end $x = L$ into a medium at 0°:

$$u(0,t) = 0, \qquad u_x(L,t) + h\,u(L,t) = 0, \quad h > 0.$$

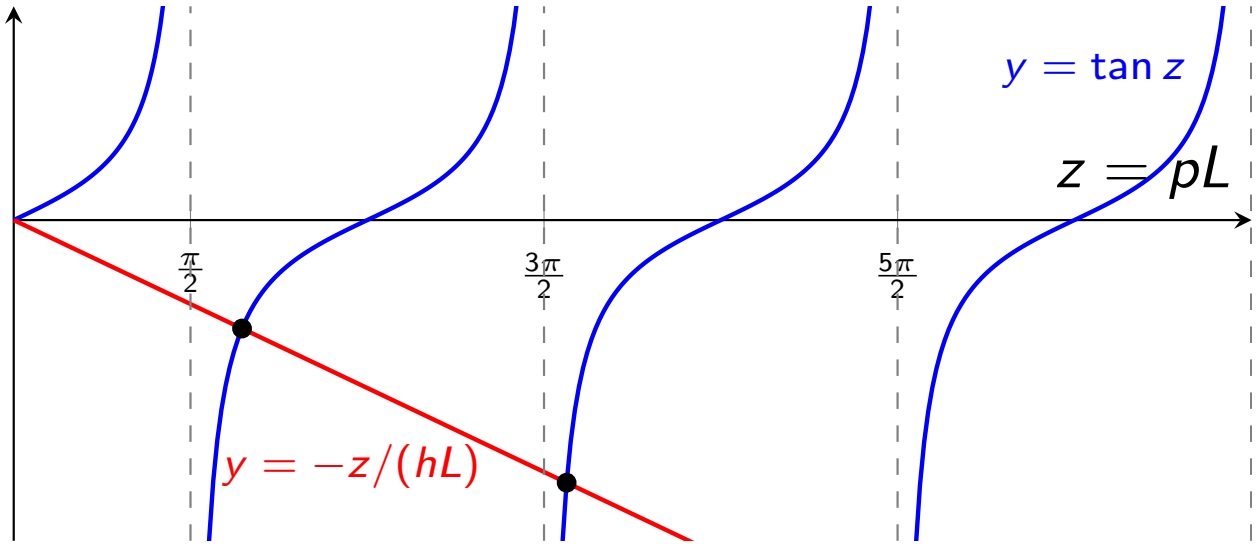
- Separation gives $X'' + p^2 X = 0$, $X(0) = 0$, $X'(L) + hX(L) = 0$ (again $\mu \geq 0$ is trivial).
- $X = \sin px$, and the end condition gives $p\cos pL + h\sin pL = 0$, i.e.

$$\tan pL = -\frac{p}{h}.$$

- This transcendental equation has infinitely many positive roots $p_1 < p_2 < \cdots$, which cannot be written in closed form.
- Solution: $u = \sum_n B_n \sin p_n x\, e^{-k^2 p_n^2 t}$ with $B_n = \dfrac{\int_0^L f \sin p_n x\, dx}{\int_0^L \sin^2 p_n x\, dx}$.



Picture: Graphical Solution of $\tan z = -z/(hL)$



- Here $hL = 1$: the roots $z_n = p_n L$ lie in $\left(\frac{(2n-1)\pi}{2}, n\pi\right)$ and $z_n \approx \frac{(2n-1)\pi}{2}$ for large n.
- Large h (cold surroundings, strong radiation) pushes the roots towards $n\pi$: the Dirichlet case.



Example 9: Bar with Lateral Heat Loss

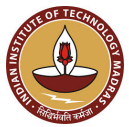
- If the lateral surface is not insulated, heat escapes at a rate proportional to u:

$$u_t = k^2 u_{xx} - hu, \quad h > 0; \qquad u(0,t) = u(L,t) = 0, \quad u(x,0) = f(x).$$

- Put $u = e^{-ht} v$. Then $u_t = e^{-ht}(v_t - hv)$ and $u_{xx} = e^{-ht} v_{xx}$.
- Substituting: $v_t - hv = k^2 v_{xx} - hv$, i.e. $v_t = k^2 v_{xx}$ with the same data.
- By Case I, $v = \sum B_n \sin \frac{n\pi x}{L} e^{-\lambda_n^2 t}$ with $B_n = \frac{2}{L} \int_0^L f \sin \frac{n\pi x}{L}\, dx$.
- Hence

$$u(x,t) = e^{-ht} \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{L}\, e^{-\lambda_n^2 t} = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{L}\, e^{-(\lambda_n^2 + h)t}.$$

- Every mode decays faster by the extra factor e^{-ht}.



Summary of the Five Cases

Boundary conditions	Solution	As $t \to \infty$
$u = 0,\ u = 0$	$\sum B_n \sin \frac{n\pi x}{L} e^{-\lambda_n^2 t}$	0
$u_x = 0,\ u_x = 0$	$A_0 + \sum A_n \cos \frac{n\pi x}{L} e^{-\lambda_n^2 t}$	average of f
$u = 0,\ u_x = 0$	$\sum B_n \sin \frac{(2n-1)\pi x}{2L} e^{-\lambda_n^2 t}$	0
$u_x = 0,\ u = 0$	$\sum A_n \cos \frac{(2n-1)\pi x}{2L} e^{-\lambda_n^2 t}$	0
$u = A_1,\ u = B_1$	$u_s + u_{tr}$	$u_s = A_1 + (B_1 - A_1)\frac{x}{L}$

- $\lambda_n = k\, p_n$, where p_n is the n-th eigenvalue of the spatial problem.
- Coefficients always come from the Fourier expansion of the initial (transient) data.



Heat Equation versus Wave Equation

- Both are solved by separation of variables with the same spatial eigenfunctions.
- Wave: $T'' = -\lambda_n^2 T$ gives oscillation $\cos \lambda_n t$, $\sin \lambda_n t$; two initial conditions needed.
- Heat: $T' = -\lambda_n^2 T$ gives decay $e^{-\lambda_n^2 t}$; only one initial condition needed.
- Wave: energy is conserved, the motion is periodic, singularities travel along characteristics.
- Heat: temperature differences are smoothed out; corners in f disappear at once.
- Wave: finite speed of propagation. Heat: a change at one point is felt everywhere instantly (infinite speed).
- Wave equation is reversible in time; heat equation is not.
- Types: wave hyperbolic, heat parabolic (recall the classification).



Additional Exercises

- (1) Solve $u_t = u_{xx}$, $0 < x < \pi$, $u(0,t) = u(\pi,t) = 0$, $u(x,0) = \sin^3 x$. (Use $\sin^3 x = \frac{3}{4}\sin x - \frac{1}{4}\sin 3x$.).
- (2) Solve $u_t = k^2 u_{xx}$, $u(0,t) = u(L,t) = 0$, $u(x,0) = x$ for $0 < x < \frac{L}{2}$ and $L - x$ for $\frac{L}{2} < x < L$.
- (3) Solve $u_t = 4u_{xx}$, $0 < x < 2$, $u_x(0,t) = u_x(2,t) = 0$, $u(x,0) = 1 + \cos \pi x + 2\cos 3\pi x$.
- (4) Solve $u_t = k^2 u_{xx} - hu$ ($h > 0$, lateral heat loss) with zero end temperatures, using $u = e^{-ht} v$.
- (5) Show that $\int_0^L u^2\, dx$ decreases in t for Dirichlet or Neumann conditions, and deduce uniqueness.
- (6) Find the first three roots of $\tan z = -z$ numerically and write the first three terms of the solution with radiation at $x = L$.

